

R Notebook

```
library(phyloseq)
library(microbiome)
```

```
## Loading required package: ggplot2
```

```
##
## microbiome R package (microbiome.github.com)
##
##
## Copyright (C) 2011-2022 Leo Lahti,
## Sudarshan Shetty et al. <microbiome.github.io>
```

```
##
## Attaching package: 'microbiome'
```

```
## The following object is masked from 'package:ggplot2':
##
## alpha
```

```
## The following object is masked from 'package:base':
##
## transform
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v lubridate  1.9.4      v tibble    3.3.0
## v purrr      1.0.4      v tidyr     1.3.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x microbiome::alpha() masks ggplot2::alpha()
## x dplyr::filter()      masks stats::filter()
## x dplyr::lag()          masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggpubr)
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
library(stringr)
library(patchwork)
library(viridis)
```

```
## Loading required package: viridisLite
```

```
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
knitr::opts_chunk$set(dev = "cairo_pdf")
```

Sequencing Summary Metrics

```
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m

seq_metrics %>%
  group_by(water_control) %>%
  summarize(Mean_reads = mean(passed_filters, na.rm = TRUE))
```

```
## # A tibble: 2 x 2
##   water_control Mean_reads
##   <chr>          <dbl>
## 1 No           3816698.
## 2 Yes          138170
```

Sample Taxon Reports

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_a
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m

# separate slurry from pond samples
pond_samples = c("s9", "s10", "s19", "s20")

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# Create a metadata dataframe to merge with taxon data
sample_metadata = seq_metrics %>%
  select(sample_name, water_control, infection_class, total_reads, subsampled_fraction, passed_filters,
  mutate(
# Library prep: MSSPE if ends with "_M"
    library_prep = if_else(str_ends(sample_name, "_M"), "MSSPE", "NGS"),

# Remove the _M suffix to get the core sample identifier
    core = str_remove(sample_name, "_M$"),
```

```

# Detect experiment 3 first (Cxxx_LPx)
is_exp3 = str_detect(core, "C\\d+.*LP\\d+"),

# Base: first sample ID (e.g. S10, PC1)
base = str_extract(core, "[A-Za-z]+\\d+"),

# Append -2 or -3 only if not exp3
suffix = case_when(
  !is_exp3 & str_detect(core, "_2$") ~ "-2",
  !is_exp3 & str_detect(core, "_3$") ~ "-3",
  TRUE ~ ""
),

base_sample = tolower(paste0(base, suffix)),

experiment = case_when(
  is_exp3 ~ 3,
  str_detect(core, "_3$") ~ 2,
  TRUE ~ 1
),
# group slurry vs environmental samples
sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry")
) %>%
select(sample_name, base_sample, total_reads, subsampled_fraction, passed_filters, library_prep, experiment)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
species_data <- merged_data %>%
  filter(tax_level == 1) %>%
  #filter(experiment == 2 | experiment == 3) %>% # filter in specific experiments
  filter(water_control == "No") %>% # filter out Negative Controls
  filter(infection_class != "Definite") # filter out positive controls

# Filter to include only species level (tax_level == 1) to avoid duplication
species_data <- merged_data %>%
  filter(tax_level == 1)

species_data_filtered = species_data %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >= 3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)

seq_metrics_tbl_1 = sample_metadata %>%
  #left_join(sample_metadata, by = "sample_name") %>%
  filter(water_control == "No") %>% # filter out Negative Controls
  filter(infection_class != "Definite") %>% # filter out positive controls
  filter(!base_sample %in% c("s5-2")) %>%
  filter(experiment == 1) %>%
  mutate(
    total_reads_sub = total_reads * subsampled_fraction # subsample total reads according to subsample
  )

```

```
seq_metrics_summary_1 = seq_metrics_tbl_1 %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters, compression_ratio) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(total_reads, passed_filters, compression_ratio),
    names_glue = "{.value}_{library_prep}"
  )
```

#Experiment 2 & 3

```
seq_metrics_tbl_2 = sample_metadata %>%
  #left_join(sample_metadata, by = "sample_name") %>%
  filter(water_control == "No") %>% # filter out Negative Controls
  filter(infection_class != "Definite") %>% # filter out positive controls
  filter(!base_sample %in% c("s5-2")) %>%
  filter(experiment == 3) %>%
  mutate(
    total_reads_sub = total_reads * subsampled_fraction # subsample total reads according to subsample
  )
```

```
seq_metrics_summary_2 = seq_metrics_tbl_2 %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters, compression_ratio) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(total_reads, passed_filters, compression_ratio),
    names_glue = "{.value}_{library_prep}"
  )
```

```
cat("\\captionsetup{labelformat=empty}")
```

Calculate total reads per sample for percentage calculations

```
sample_totals <- species_data %>%
  group_by(sample_name, base_sample, library_prep) %>%
  summarise(
    total_nt_count = sum(nt_count, na.rm = TRUE),
    total_nr_count = sum(nr_count, na.rm = TRUE),
    total_nt_rpm = sum(nt_rpm, na.rm = TRUE),
    total_nr_rpm = sum(nr_rpm, na.rm = TRUE),
    .groups = "drop"
  )
```

Extract viral category data

```
virus_data <- species_data %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false") # filter out phages
```

Calculate percentage of viral reads per sample

```
viral_percentages <- virus_data %>%
  drop_na(base_sample) %>%
  group_by(sample_name, base_sample, library_prep) %>%
  summarise(
    viral_nt_count = sum(nt_count, na.rm = TRUE),
```

```

    .groups = "drop"
  ) %>%
  left_join(sample_totals, by = c("base_sample", "library_prep")) %>%
  mutate(
    viral_nt_percent = (viral_nt_count / total_nt_count) * 100,
    # Replace NaN and Inf with NA
    viral_nt_percent = ifelse(is.nan(viral_nt_percent) | is.infinite(viral_nt_percent), NA, viral_nt_percent)
  )

viral_summary = viral_percentages %>%
  select(base_sample, library_prep, viral_nt_count, viral_nt_percent) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(viral_nt_count, viral_nt_percent),
    names_glue = "{.value}_{library_prep}"
  )

# join viral calculations to summary table
seq_metrics_summary_1 = seq_metrics_summary_1 %>%
  left_join(viral_summary, by = "base_sample", "library_prep")
seq_metrics_summary_1

```

A tibble: 8 x 11

```

base_sample total_reads_MSSPE total_reads_NGS passed_filters_MSSPE 1 s6 7722606 10097840
2998354 2 s9 3502792 24845878 2842370 3 s5 15342574 7517708 3661612 4 s7 1273898 518666 609466 5
s2 2221908 455222 1825676 6 s4 1204728 1001116 1018914 7 s8 4303916 1199608 2720762 8 s10 54021306
8850592 5581604 # i 7 more variables: passed_filters_NGS , compression_ratio_MSSPE , # compression_ratio_NGS , viral_nt_count_NGS , # viral_nt_count_MSSPE , viral_nt_percent_NGS , #
viral_nt_percent_MSSPE

```

```

# Make table pretty
seq_metrics_summary_1 %>%
  mutate(
    passed_filters_msspe = (passed_filters_MSSPE / total_reads_MSSPE) * 100,
    passed_filters_ngs = (passed_filters_NGS / total_reads_NGS) * 100
  ) %>%
  select(base_sample, total_reads_NGS, total_reads_MSSPE, passed_filters_ngs, passed_filters_msspe, compression_ratio_NGS, compression_ratio_MSSPE, viral_nt_percent_NGS, viral_nt_percent_MSSPE) %>%
  arrange(desc(viral_nt_percent_MSSPE)) %>%
  rename(
    "Sample" = base_sample,
    "Reads (MSSPE)" = total_reads_MSSPE,
    "Reads (NGS)" = total_reads_NGS,
    "Passed Filter % (MSSPE)" = passed_filters_msspe,
    "Passed Filter % (NGS)" = passed_filters_ngs,
    "DCR (NGS)" = compression_ratio_NGS,
    "DCR (MSSPE)" = compression_ratio_MSSPE,
    "Viral Reads % (MSSPE)" = viral_nt_percent_MSSPE,
    "Viral Reads % (NGS)" = viral_nt_percent_NGS
  ) %>%
  kable(
    format = "latex",
    caption = "Experiment 1: Shallow Sequencing",
  )

```

Experiment 1: Shallow Sequencing

Sample	Reads (NGS)	Reads (MSSPE)	Passed Filter % (NGS)	Passed Filter % (MSSPE)	DCR (NGS)	DCR (MSSPE)	Viral Reads % (NGS)	Viral Reads % (MSSPE)
s7	518,666	1,273,898	40.78	47.84	1.31	1.32	10.81	6.16
s8	1,199,608	4,303,916	59.53	63.22	1.41	1.49	5.90	5.70
s4	1,001,116	1,204,728	66.06	84.58	1.38	1.36	3.50	3.23
s6	10,097,840	7,722,606	35.73	38.83	1.79	1.50	1.46	2.47
s2	455,222	2,221,908	50.17	82.17	1.33	1.43	2.42	1.55
s5	7,517,708	15,342,574	49.46	23.87	1.85	1.84	1.21	1.33
s9	24,845,878	3,502,792	16.82	81.15	2.10	1.49	0.43	0.45
s10	8,850,592	54,021,306	46.24	10.33	2.06	2.77	0.16	0.11

```

    digits = 2,
    format.args = list(big.mark = ",",")
) %>%
kable_styling(
  full_width = TRUE,
  position = "center",
)

```

```

# join viral calculations to summary table
seq_metrics_summary_2 = seq_metrics_summary_2 %>%
  left_join(viral_summary, by = "base_sample","library_prep")

seq_metrics_summary_2 %>%
  mutate(
    passed_filters_msspe = (passed_filters_MSSPE / total_reads_MSSPE) * 100,
    passed_filters_ngs = (passed_filters_NGS / total_reads_NGS) * 100
  ) %>%
  select(base_sample, total_reads_NGS, total_reads_MSSPE, passed_filters_ngs, passed_filters_msspe, comp
  arrange(desc(viral_nt_percent_MSSPE)) %>%
  rename(
    "Sample" = base_sample,
    "Reads (MSSPE)" = total_reads_MSSPE,
    "Reads (NGS)" = total_reads_NGS,
    "Passed Filter % (MSSPE)" = passed_filters_msspe,
    "Passed Filter % (NGS)" = passed_filters_ngs,
    "DCR (NGS)" = compression_ratio_NGS,
    "DCR (MSSPE)" = compression_ratio_MSSPE,
    "Viral Reads % (MSSPE)" = viral_nt_percent_MSSPE,
    "Viral Reads % (NGS)" = viral_nt_percent_NGS
  ) %>%
  kable(
    format = "latex",
    caption = "Experiment 2: Deep Sequencing",
    digits = 2,
    format.args = list(big.mark = ",",")
  ) %>%
  kable_styling(

```

Experiment 2: Deep Sequencing

Sample	Reads (NGS)	Reads (MSSPE)	Passed Filter % (NGS)	Passed Filter % (MSSPE)	DCR (NGS)	DCR (MSSPE)	Viral Reads % (NGS)	Viral Reads % (MSSPE)
s11	51,727,246	50,818,782	6.79	6.28	1.76	1.59	6.06	10.82
s17	31,579,706	47,462,476	27.86	10.15	4.41	2.42	8.53	9.86
s15	38,795,268	93,010,162	18.81	11.12	3.64	5.17	7.07	9.35
s12	55,612,598	45,556,186	7.34	8.51	2.03	1.94	5.96	6.56
s16	31,110,126	54,777,360	7.94	4.70	1.24	1.29	1.28	3.90
s20	24,564,082	79,608,770	15.66	4.01	1.92	1.60	1.26	3.67
s13	43,297,796	46,082,456	10.33	9.82	2.23	2.28	3.55	3.39
s14	45,011,752	61,293,528	7.64	7.36	1.72	2.25	2.91	3.06
s19	81,398,780	75,310,306	3.59	3.74	1.46	1.41	2.44	2.62
s18	56,664,844	55,901,566	6.81	5.01	1.93	1.40	1.09	2.14

```

full_width = TRUE,
position = "center",
)

```

Wilcoxon signed rank test to measure differences between library preps

#seq metrics ALL

```

seq_metrics_tbl = sample_metadata %>%
  filter(water_control == "No") %>% # filter out Negative Controls
  filter(infection_class != "Definite") %>% # filter out positive controls
  filter(!base_sample %in% c("s5-2")) %>%
  mutate(
    total_reads_sub = total_reads * subsampled_fraction # subsample total reads according to subsample
  )

seq_metrics_summary = seq_metrics_tbl %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(total_reads, passed_filters),
    names_glue = "{.value}_{library_prep}"
  )

seq_metrics_tbl %>%
  group_by(library_prep) %>%
  summarize(mean_reads = mean(total_reads),
    mean_reads_sub = mean(total_reads_sub))

```

```

## # A tibble: 2 x 3
##   library_prep mean_reads mean_reads_sub
##   <chr>         <dbl>         <dbl>
## 1 MSSPE         43212083.        5303578.
## 2 NGS           28569379.        3775220.

```

```

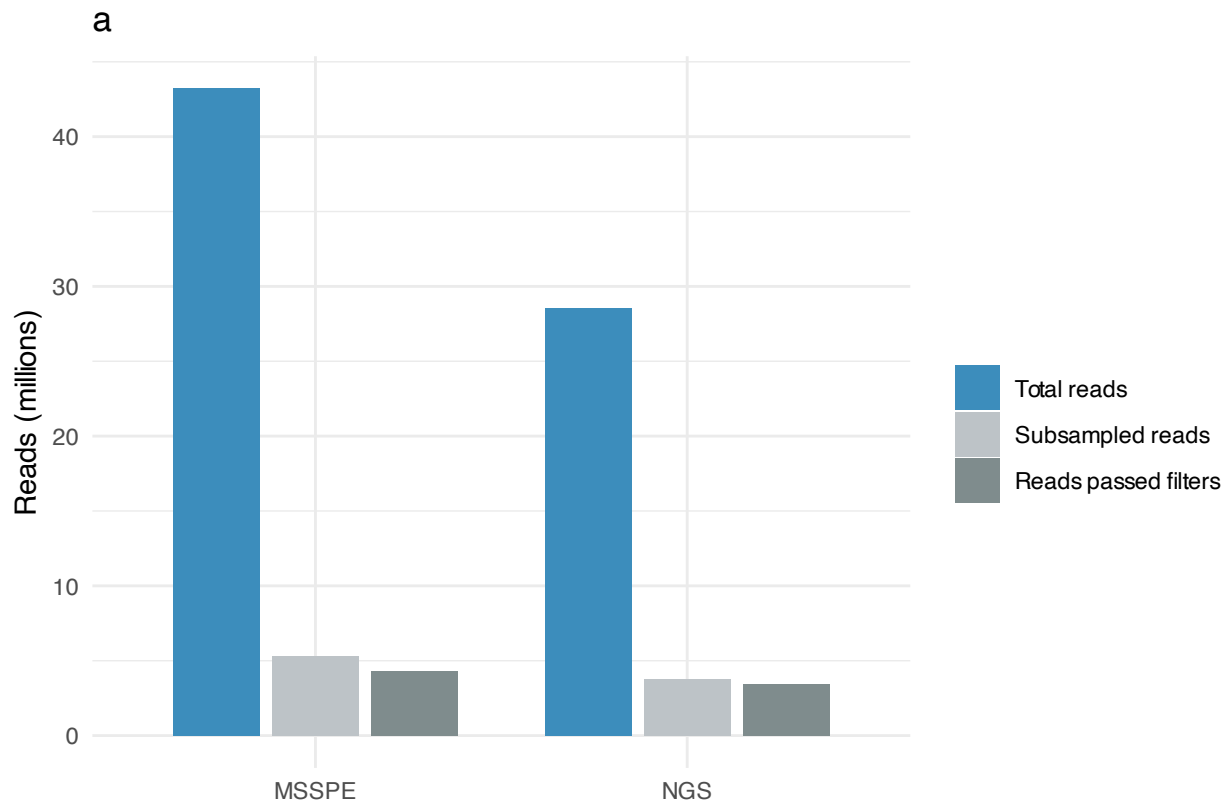
seq_metrics_tbl_long <- seq_metrics_tbl %>%
  pivot_longer(
    cols = c(total_reads, total_reads_sub, passed_filters),
    names_to = "metric",
    values_to = "value"
  ) %>%
  mutate(metric = factor(metric, levels = c("total_reads", "total_reads_sub", "passed_filters")),
         value = value / 1e6,
         metric = recode(
           metric,
           "total_reads" = "Total reads",
           "total_reads_sub" = "Subsampled reads",
           "passed_filters" = "Reads passed filters"
         )
  )

plot_data <- seq_metrics_tbl_long %>%
  group_by(library_prep, metric) %>%
  summarize(mean_value = mean(value), .groups = "drop")

overall_barplot = ggplot(plot_data, aes(x = library_prep, y = mean_value, fill = metric)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7, linewidth = 0.2) +
  scale_fill_manual(
    values = c(
      "Total reads" = "#3C8DBC",    # professional blue
      "Reads passed filters" = "#7F8C8D", # neutral gray
      "Subsampled reads" = "#BDC3C7"    # light gray for secondary metric
    ),
    name = ""
  ) +
  labs(
    title = "a",
    x = "",
    y = "Reads (millions)"
  ) +
  theme_minimal()

overall_barplot

```



```
library(ggnewscale)
library(RColorBrewer)

# Create boxplot for viral NT percentage by control status
mean_df_viral_perc <- viral_percentages %>%
  group_by(library_prep) %>%
  summarise(mean_value = mean(viral_nt_percent, na.rm = TRUE))

viral_summary_wilcox = viral_summary %>%
  mutate(
    viral_nt_percent_NGS = replace_na(viral_nt_percent_NGS, 0), # set NAs to 0 for non detections
    viral_nt_percent_MSSPE = replace_na(viral_nt_percent_MSSPE, 0), # set NAs to 0 for non detections
  )

# Paired Wilcoxon test across all samples
wilcox_viral_perc <- wilcox.test(
  viral_summary_wilcox$viral_nt_percent_NGS,
  viral_summary_wilcox$viral_nt_percent_MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)

## Warning in wilcox.test.default(viral_summary_wilcox$viral_nt_percent_NGS, :
## cannot compute exact p-value with zeroes

pval_viral_perc <- wilcox_viral_perc$p.value
```

```

# get stupid color schemes so I cant make it continuous
blues_palette <- brewer.pal(9, "Blues") # Get a blues palette with 9 colors
reds_palette <- brewer.pal(9, "Reds") # Get a reds palette with 9 colors

library_prep_levels <- levels(factor(viral_percentages$library_prep))

viral_nt_boxplot <- ggplot(viral_percentages, aes(x = library_prep, y = viral_nt_percent)) +
  geom_boxplot(outlier.fill = "white", outlier.alpha = 0.5) +
  geom_jitter(
    data = subset(viral_percentages, library_prep == "MSSPE"),
    aes(color = viral_nt_percent),
    width = 0.1, alpha = 0.7, size = 1.8
  ) +
  scale_color_gradientn(colors = blues_palette) +
  #scale_color_viridis(option = "plasma", direction = -1, limits = c(0, max(viral_percentages$viral_nt_perce
  new_scale_color() +
  geom_jitter(
    data = subset(viral_percentages, library_prep == "NGS"),
    aes(color = viral_nt_percent),
    width = 0.1, alpha = 0.7, size = 1.8
  ) +
  scale_color_gradientn(colors = reds_palette) +
  #scale_color_viridis(option = "D", direction = -1, limits = c(0, max(viral_percentages$viral_nt_perce
  annotate("text", x = mean(1:length(library_prep_levels)), y = max(viral_percentages$viral_nt_perce,
    label = paste0("Wilcoxon p = ", signif(pval_viral_perc, 3)),
    fontface = "italic", size = 4) +
  stat_summary(fun = median, geom = "text", aes(label = round(..y.., 1)),
    vjust = -0.5, color = "black", fontface = "bold", size = 3) +
  geom_segment(data = mean_df_viral_perc,
    aes(x = as.numeric(library_prep) - 0.3,
        xend = as.numeric(library_prep) + 0.3,
        y = mean_value, yend = mean_value),
    linetype = "dotted", color = "black") +
  geom_text(data = mean_df_viral_perc,
    aes(x = library_prep, y = mean_value, label = paste0(": ", round(mean_value, 1))),
    vjust = -1, color = "black", size = 3.2, fontface = "italic") +
  theme_bw() +
  labs(
    title = "b",
    x = "",
    y = "% Viral Reads"
  ) +
  theme(
    #axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )

viral_nt_boxplot

```

```

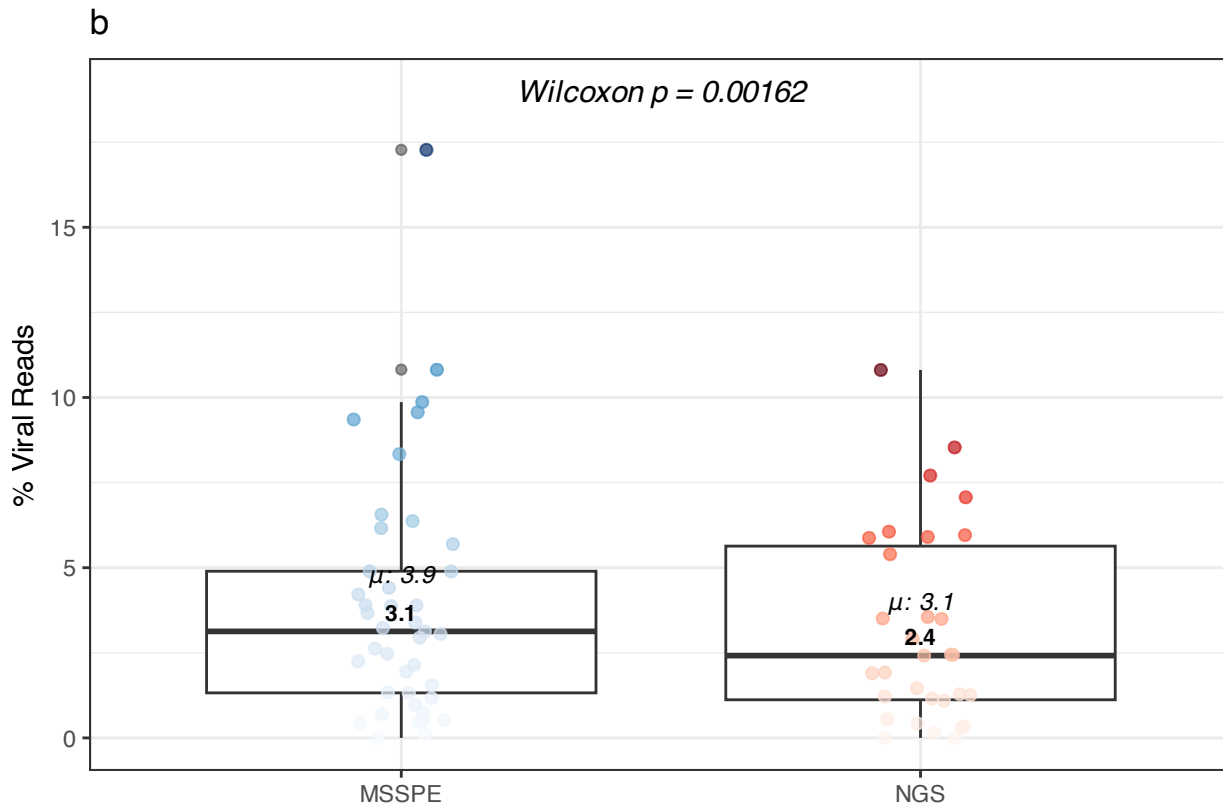
## Warning: The dot-dot notation (`..y..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(y)` instead.

```

```
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_segment()`).
```



```
horizontal_space <- plot_spacer() + plot_layout(widths = c(0.2))

combined_boxplots = (overall_barplot | horizontal_space | viral_nt_boxplot) +
  plot_layout(ncol = 3, widths = c(1.5, 0.5, 1.5))

combined_boxplots
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_segment()`).
```

